

Best Buddies: The Web and Open Source Electronics

Here are a few interesting online locations that help you learn about, use, build and share open source electronics



Tabby, an open source universal chassis that can be assembled in less than an hour (Courtesy: OSVehicle)

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Although the World Wide Web (WWW) itself is considered by many as the biggest success of open source (OS), the dependence of OS on the WWW is actually even greater. Imagine how difficult it would be to create, share and collaborate with people across the world without the Web as a backbone? An issue focusing on OS electronics would therefore be quite incomplete if we do not talk about online resources that aid in the development and use of such electronics. Here, we look at some such interesting sites that satiate your cu-

riosity and enthusiasm to build open hardware. However, note that we are not repeating the oft-mentioned and already well-known resources such as Arduino or Raspberry Pi's official blogs, Make Magazine, and such.

1. How can I get my PCB designs fabbed?

There are several OS tools available on the Web to design printed circuit boards (PCB). However, many often think it is futile to fiddle with designs if they cannot get them fabricated. Now, the Web also makes such facilities available at your doorstep.

During his I-Day speech, our Prime Minister insisted on manufacturing in India. He invited people to make in

India, sell anywhere in the world. So, when it comes to your PCBs too, made in India ought to be the first choice to explore. PCB Power (www.pcbpower.com) is an Indian firm specialising in design and manufacture of PCBs. While they supply to customers in hi-fi verticals like aviation, defence, railways, industrial electronics, etc, what sets them apart from others is that they also print small orders for learners and enthusiasts! Ordering is quite easy and unambiguous too. First, enter basic specifications like size, layers, delivery period and quantity, and get a quick online quote. If you are okay with the rates, you can place an order by uploading the Gerber files and fabrication parameters.

Another choice is German manufacturer Fritzing Fab (<http://fab.fritzing.org/fritzing-fab>), which manufactures in Berlin and supplies to any place in the world. Fritzing Fab consolidates orders and fabricates them together once a week, which makes it possible to process even single-PCB orders at a reasonably low cost. Apart from fabrication services, Fritzing also offers loads of learning materials, and a freely downloadable OS software that helps you create, document and share OS electronics.

OSH Park (<https://oshpark.com/>) is another option, where PCB orders from the community are consolidated, manufactured in the USA, and shipped for free to anywhere in the world. Again, this is an affordable option, and OSH Park ships promptly within 12 to 15 days of ordering. To get an idea, they charge US\$ 5 per square inch for three copies of two-layer boards, and